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MILTARY SPECIFICATION

CONTROL CABLE ASSEMBLIES AND CONNECTOR PLUGS AND RECEPTACLES (FOR TORPEDOES)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement, manufacture, and inspection of torpedo fire control multiconductor cable assemblies and connector plugs and receptacles.

1.2 Classification. Cable assemblies and associated components shall be of the following types and classes, as specified on Drawing 783794.

1.2.1 Types

(a) Type I: Cable assemblies and connector plugs and receptacles subject to sea pressure (700 psi).

(b) Type II: Cable assemblies and connector plugs and receptacles subject to pneumatic pressure (250 psi).

(c) Type III: Cable assemblies and connector plugs and receptacles subject to pressures encountered under shipboard conditions (25 psi).

1.2.2 Classes

(a) Class 1: 65-conductor "A" cable assemblies and connector plugs and receptacles.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer Naval Ordnance Station, Standardization Division (Code 611). Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(b) Class 2: 65-conductor "B" cable assemblies and connector plugs and receptacles for submarine torpedo tubes.

(c) Class 3: 65-conductor "B" cable assemblies and connector plugs and receptacles for surface vessel torpedo tubes.

(d) Class 4: 65-conductor "C" cable assemblies and connector plugs and receptacles.

(e) Class 5: 4- and 10-conductor cable connector assemblies (plugs and receptacles).

(f) Class 6: Stowage rack and test cable connector assemblies (plugs and receptacles).

1.2.3 Service ratings

NOMINAL WIRE SIZE		NOMINAL CURRENT RATING			
Class 1	Classes 2, 3, 4	Amps	Service	DC Volts	AC Volts (rms)
24	22	1.00	Instrument	24	---
20	18	2.30	Single Phase Power	--	115
18	16	5.0	Service Power Firing Circuit	--	115
18	16	12.0	Service Power DC	24	---

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, the requirements of this specification shall apply.

SPECIFICATIONS

Federal

PPP-B-585 Boxes, Wood, Wirebound

PPP-B-591 Boxes, Fiberboard, Wood-Cleated

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PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-S-901	Shipboard Equipment-High Impact Shock Tests
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

Federal

FED-STD-406	Plastics: Methods of Testing

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-810	Environmental Test Methods

DRAWINGS

Naval Sea Systems Command (Code Ident 10001)

LD 542923	Torpedo Connector Receptacle
Drawing 1515850	Torpedo Control Cable Mk 1 Mod 13
Drawing 783794	Torpedo Control Cables and Associated Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposals shall supply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description

3.1.1 Torpedo control cable assemblies. As used herein, torpedo control cable assemblies shall comprise the 65-conductor cable, complete with plug or connector units, as required, assembled on each end of a specific length of cable.

3.1.2 Cable connector assemblies. As used herein, torpedo cable connector assemblies shall comprise the assembled parts of the connector plugs and receptacles as a unit assembly, without cable, representing a unit which will assume its function at the end of the torpedo control cable assembly, as required in 3.1.1, or as a complete unit for use as a stock spare item.

3.1.3 Classification of connector plug and receptacle assemblies. When used as part of a complete torpedo control cable assembly, the connector assemblies shall assume the classification under which the complete cable assembly is identified, as specified in 1.2.

3.2 General requirements

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of torpedo fire control cable assemblies and connector plugs and receptacles shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing units that comply with the technical requirements of the contract. See 4.3.1.

3.2.2 Design and construction. Cable assemblies and connector plug and receptacle assemblies and parts covered by this specification shall be manufactured to conform to the applicable drawings referenced on Drawing 783794.

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3.2.3 Periodic production sample. Unless otherwise specified in the contract or order torpedo fire control cable assemblies and connector plug and receptacle assemblies shall be subjected to and successfully pass the periodic production inspection and test requirements of 4.3.3.

3.2.4 Action in case of failure. In case of failure of any cable assembly or connector plug or receptacle assembly to pass specific inspections or tests as stated above, the failure actions stated in 4.4 shall be initiated.

3.2.5 Quality conformance inspection. Unless otherwise specified in the contract or order, torpedo fire control cable assemblies and connector plug and receptacle assemblies shall be subjected to and successfully pass the quality conformance tests and examinations of 4.5.

3.2.6 Materials. Unless otherwise specified in the contract or order, all materials and purchased parts entering into the final assembly of torpedo fire control cable and connector plugs and receptacles shall conform strictly to the detailed requirements indicated on the drawings and specifications referenced on Drawing 783794.

3.2.6.1 Special materials. Materials which are not covered by applicable specifications, or which are not specifically described herein, shall be of the best quality and be such as to permit the cable and associated components to meet the functional and environmental requirements and tests specified herein. Prior approval for use of these materials in production shall be obtained from the cognizant technical authority at the Naval Sea Systems Command.

3.2.7 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for torpedo fire control cable assemblies and connector plugs and receptacles shall be strictly in accordance with the applicable drawings, specifications, and publications referenced on Drawing 783794.

3.2.8 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with applicable drawings and standards listed on Drawing 783794.

3.2.9 Interchangeability. Fabrication of component parts shall be such that size, fit, and assembly of component parts assure proper fit and interchangeability without interference, binding, or misalignment and without resorting to selection or fitting. When required by design, interference fits are, or shall be, detailed on the applicable drawings.

3.2.10 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

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3.2.11 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Performance requirements and product characteristics

3.3.1 Orientation. Dimensional stability shall be maintained so that when contact insert assemblies are integral with the final assembly housings, orientation of contacts and pins and mating housing diameters shall permit ease of assembly of applicable mating parts and full mating and unmating by hand without the use of tools, or when tested as in Section 4.

3.3.2 Product characteristics. There are 65 channels available to fire control through the conductor circuits. The designs referenced on Drawing 783794 are standardized for use in conjunction with present fire control systems.

3.3.3 Mechanical

3.3.3.1 Mating or unmating force. All 65-contact connector plugs and receptacles, when subjected to tests as defined in 4.5.3.3 and 4.5.3.4 shall be capable of engaging with mating assemblies referenced on Drawing 783794 under a force not exceeding 52 pounds. Unmating force shall be not less than 20 pounds nor more than 52 pounds.

3.3.3.2 Cable separation force. Type I Class 1 cable assemblies shall be so constructed as to permit, after assembly with the torpedo connector receptacle LD 542923, separation by cable cutter of the cable from the torpedo connector plug upon application of a force in tension of not less than 50 nor more than 70 pounds as specified in 4.6.4.

3.3.3.3 Cable entrance pull test. Each cable assembly, connector plug, and receptacle shall withstand the 100 $\pm$ 2 pound cable entrance pull test as specified in 4.5.3.2.

3.3.4 Electrical

3.3.4.1 Continuity. All assemblies shall be tested for continuity. All contacts shall evidence continuity when tested in accordance with 4.5.3.5.1. This test shall be applied to each contact-conductor circuit in the assembled state as specified in 4.5.3.5.1.

3.3.4.2 Contact resistance. Insert contact assemblies shall have a contact resistance of 0.5 ohm maximum when tested in accordance with 4.5.3.5.2.

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3.3.4.3 Dielectric strength. The insert material noted on drawings referenced on Drawing 783794 shall meet a dielectric strength of 325 volts per mil without material breakdown when tested in accordance with FED-STD-406 Method 4031, as specified in 4.6.3.

3.3.4.4 Dielectric withstanding voltage (hipot) and insulation resistance. Leakage current shall not exceed 500 microamperes when tested in accordance with 4.5.3.5.3.1. Insulation resistance shall not be less than one megohm when tested in accordance with 4.5.3.5.3.2. In addition, for Classes 1, 2, and 3 torpedo control cable final assemblies, the electrical tests shall be performed while the assemblies are being subjected to hydrostatic tests as specified in 4.5.3.6.

3.4 Environmental conditions. Torpedo control cable assemblies and connector plugs and receptacles shall be capable of meeting the performance requirements and product characteristics specified herein after or while being subjected to the following, when tested as specified in Section 4:

3.4.1 Vibration

3.4.1.1 Vibration (nonoperating). The control cable assemblies and connector assemblies, all types, when engaged with the mating connectors or cables, shall meet the performance requirements and product characteristics of section 3.3 after being subjected to the test specified in 4.6.1.1.

3.4.1.2 Vibration (transportation). The control cable assemblies and connector assemblies, after being subjected to the test specified in 4.6.1.2, shall show no damage and shall meet the performance requirements and product characteristics of Section 3.3.

3.4.2 Shock

3.4.2.1 Shock (nonoperating). The control cable assemblies and connector assemblies, when engaged with the mating connectors or cables, shall meet the performance requirements and product characteristics of Section 3.3 after being subjected to the test required for lightweight equipment in accordance with MIL-S-901, Grade A, Class I as specified in 4.6.1.3.

3.4.3 Temperature and humidity. Torpedo connector receptacle assemblies and torpedo control cable assemblies shall meet the performance requirements and product characteristics of Section 3.3 when subjected to the temperature and humidity tests specified in 4.6.1.4.

3.4.4 Pressure

3.4.4.1 Cables and connectors (Type I). Cable assemblies and connector plugs and receptacles shall meet the performance requirements and product

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characteristics as specified in Section 3.3 after being subjected to the 500 and 700 psi hydrostatic pressure cycle tests as specified in 4.5.3.6.1.

3.4.4.1.1 Breech door plug (Type I Class 1)(Following cable cutter operation). Breech door connector plug and the cable length remaining after cable cutter operation shall withstand the 600 ± 20 psi hydrostatic pressure test of 4.6.5.

3.4.4.2 Receptacles, Type I (sea exposure side). Receptacle assemblies shall meet the performance requirements and product characteristics of Section 3.3 after being subjected (externally) to a hydrostatic pressure of 700 psi for a time duration of 15 minutes, after which a hydrostatic test of 5 psi shall be performed, as specified in 4.5.3.6.2.

3.4.4.3 Receptacles Type II. Type II Class 3 receptacle assemblies shall meet the performance requirements and product characteristics of Section 3.3 after being subjected (externally) to a hydrostatic pressure of 250 psi, as specified in 4.6.2.1.

3.4.4.4 Pressures (shipboard conditions). Type III (all classes) connector assemblies and torpedo control cable assemblies shall meet the performance requirements and product characteristics of Section 3.3 after being subjected (externally) to 25 psi hydrostatic pressure for a duration of 10 minutes, as specified in 4.6.2.2.

3.4.5 Marine environment. The product shall function in accordance with performance requirements of Section 3.3 after exposure to natural marine environment under the following conditions: Product assembled unpackaged but open to salt laden atmosphere for 96 hours as specified in 4.6.1.5.

3.5 Cleanup (final assembly). After final assembly all cable assemblies and connector assemblies and components shall be free of loose, spattered or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed (i.e., all material protuberances likely to foster chips or broken material in product performance).

3.6 Workmanship. All products under this specification, including components, shall be constructed and finished in such a manner as to meet all the requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation, marking of parts and assemblies, plating, protective coatings, machine screw assemblage, welding, brazing, or any mechanical protuberances depressing the quality of material or interfering with the performance of the product. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the control cables and associated components shall be classified as follows:

(a) Preproduction inspection (paragraph 4.6)

(b) Quality conformance inspection (paragraph 4.5)

(c) Periodic production inspection (paragraph 4.6)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of five torpedo fire control cables and connector plugs and receptacles, each manufactured in accordance with 3.2.1, shall be subjected to the preproduction tests detailed in 4.6 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the product by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production sample. From each succeeding lot or grand lot of 300 that have passed the quality conformance inspection of 4.5, a random sample of five torpedo control cable assemblies and connector plugs and receptacles will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The torpedo control cable assemblies and connector plugs and receptacles shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.5 and 4.6. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

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4.4 Action in case of failure

4.4.1 Preproduction Tests. Failure of a preproduction sample to successfully pass any of the required tests of 4.3.1 shall be cause for rejection. The contractor shall determine the cause of failure and the necessary hardware corrections. When specified by the contract or purchase order, a statement of cause of failure and proposed corrective action shall be submitted to the procuring activity for approval before hardware corrections are made. Any or all preproduction tests shall then be repeated at the discretion of the procuring activity.

4.4.2 Periodic production tests. Failure of a periodic production sample to pass any of the required inspections and tests of 4.3.3 shall be cause for rejection of that sample and its respective lot. Final acceptance of all units of the applicable lot shall be withheld pending successful completion of the following action by the contractor:

(a) Determine cause of defect.

(b) Submit to the procuring activity for approval the proposed corrective action to prevent recurrence of the defect.

(c) Upon approval, make hardware corrections to sample or samples and repeat any or all periodic production tests at the discretion of the procuring activity.

(d) Take necessary action to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(e) Periodic production sampling on subsequent production shall revert to 4.3.3.

4.5 Quality conformance inspection.

4.5.1 Each torpedo fire control cable assembly and connector plug and receptacle delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.2 Test conditions. Unless otherwise specified, the torpedo fire control cable assemblies and connector plugs and receptacles shall be subjected to quality conformance tests under the following conditions:

4.5.2.1 Temperature. Room ambient 65 degrees Fahrenheit (F) to 95 degrees F (18 degrees centigrade (C) to 35 degrees C).

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4.5.2.2 Altitude. Normal ground atmospheric pressure.

4.5.2.3 Vibration. None.

4.5.2.4 Humidity. Room ambient to 95 percent relative maximum (except for the dielectric test, for which specific humidity is delineated in 3.3.4.3).

4.5.3 Test procedure

4.5.3.1 Visual inspection. Visual inspection shall be made on each unit to determine compliance with the requirements of this specification, drawings, and applicable specifications with respect to materials, workmanship, and marking.

4.5.3.1.1 Workmanship. Visual inspection affecting all parts and accessories shall cover neatness, thoroughness of soldering, plating, anodizing, accuracy of dimensions, and proper marking of parts and assemblies. Particular attention shall be paid to removal of excess solder and other foreign material during and after final assembly.

4.5.3.2 Cable entrance pull test; torpedo control cable assemblies and connector plugs and receptacles. Each assembly shall be mounted in a fixture which simulates the actual mated coupling in service installation (figure 1 illustrates test setup for Type I Class 1 cable). A tensile force of 100 $\pm$ 2 pounds shall be exerted along the longitudinal axis of the cable connector housing entrance port. At this point the product shall be considered acceptable if there is no physical movement of the cable in the direction of the tensile force. When the connector plug and receptacle assemblies are furnished without cable, applicable cable in appropriate lengths shall be used simulating the exact locking features as required. Any permanent deviation (drift) from the applicable assembly drawing dimensions affecting the functional requirements of the cable shall be cause for rejection. The continuity test of 4.5.3.5.1 shall be performed following this pull test.

4.5.3.3 Mating force (contact assembly). All 65-contact connector plugs and receptacles shall be set into a device for simulating coupling assembly condition in service use (see figure 2). Mating plugs and receptacles shall be assembled with an axial compressive force of 52 pounds maximum. These units shall bottom when fully mated. Forces required beyond this limit shall constitute sufficient cause for rejection.

4.5.3.4 Unmating force (contact assembly). The 65-contact connector plugs and receptacles mated in 4.5.3.3 shall be completely uncoupled when an axial tensional force of 20 to 52 pounds is applied. Forces outside these limits shall constitute sufficient cause for rejection.

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4.5.3.5 Electrical tests

4.5.3.5.1 Continuity. Completed cable assemblies shall be placed in a test fixture and each contact conductor circuit checked for continuity (see Figure 3). A resistance greater than 0.5 ohm shall be considered cause for rejection.

4.5.3.5.2 Contact resistance. Insert-contact parts, when assembled in connectors (plug and receptacle), shall be mounted in a test fixture (Figure 4 illustrates test setup for type I class 1 cable). Each contact shall then be subjected to its rated voltage and current. Contact resistance shall meet the maximum resistance requirements specified in 3.3.4.2.

4.5.3.5.3 Dielectric withstanding voltage (hipot) and insulation resistance tests. These tests apply to all torpedo connector plug and receptacle assemblies and torpedo control cable assemblies. Failure to meet test requirements shall be cause for rejection.

4.5.3.5.3.1 Dielectric withstanding voltage (hipot). There shall be not more than 500 microamperes leakage current when 515±15 volts ac (rms) or 725±25 volts dc is applied for 7.85 to 9.00 seconds between each conductor and the remaining 64 conductors (with the exception of pin L) and the connector shell. Each conductor shall be tested sequentially in rotation in accordance with MIL-STD-202, Method 301, for dielectric withstanding voltage tests.

4.5.3.5.3.2 Insulation resistance. This test shall be performed immediately following the dielectric withstanding voltage test (4.5.3.5.3.1). The insulation resistance shall be not less than 1 megohm when subjected to 725±25 volts dc potential between each conductor and the remaining 64 conductors (with the exception of pin L) and connector shell. Each conductor shall be tested sequentially in rotation in accordance with MIL-STD-202, Method 302, for insulation resistance tests.

4.5.3.6 Pressure hydrostatic

4.5.3.6.1 Pressure 700 psi (external). All torpedo control cable assemblies Type I Class 1 shall be mounted in a fixture simulating, as in service use, the tube door connector water seal and torpedo receptacle seal (see Figure 5). The assembly shall then be subjected to a sustained hydrostatic pressure of 500 psi ±20 psi for a time duration of 15 minutes minimum. The pressure shall then be completely relieved and brought back to 5 psi ± 1 psi for a period of 1 minute minimum. The pressure shall be raised to 500 psi for three more 15 minute cycles with a period of one minute minimum at

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1 ± 1 psi between cycles. The pressure shall then be raised to 700 psi ± 20 psi for time duration of 15 minutes minimum. The electrical tests specified in 4.5.3.5.1 and 4.5.3.5.3 shall be conducted during the latter part of this period. Electrical breakdown, discontinuity, evidence of leakage past water seals, mechanical breakage, or permanent distortion of internal components shall be cause for rejection.

4.5.3.6.2 Pressure 700 psi (external, sea side). Torpedo connector receptacle Type I Class 1 and submarine torpedo tube door receptacle, connector Type I Class 2, shall be mounted in fixtures simulating, as in service use, the torpedo receptacle water seal and the tube door connector water seal respectively (see Figure 5). These shall then be subjected to a sustained hydrostatic pressure of 700 psi ± 20 psi for a time duration of 15 minutes minimum. The pressure shall then be completely relieved and brought back to 5 psi ± 1 psi for a period of one minute minimum. Evidence of leakage past water seals, mechanical breakage, or permanent distortion of internal components shall be cause for rejection.

4.5.3.7 Packaging, packing, and marking. Packaging, packing, and marking shall be inspected for compliance with the requirements of Section 5.

4.6 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5 shall be subjected to the following tests and examinations. When specified in the contract or purchase order, certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the Naval Sea Systems Command (see 6.2).

4.6.1 Environmental tests

4.6.1.1 Vibration (nonoperating). Torpedo control cables and connector plugs and receptacles shall be rigidly mounted to a vibration fixture so as to simulate (coupled with the mating unit) the service installation. The vibration fixture in turn shall be mounted to a sinusoidal vibration system as illustrated in Figure 6. Test units shall conform to applicable drawings referenced on Drawing 783794. The test unit shall be vibrated in accordance with the following schedule in each of its three mutually perpendicular axes. When conducting the exploratory vibration test the change in frequency shall be made in discrete frequency intervals of 1 HZ and maintained at each frequency for not less than 15 seconds. The frequency at which resonance occurs shall be noted. During the variable frequency test the change in frequency of vibration shall be made in discrete frequency intervals of 1 HZ and maintained at each frequency for 5 minutes.

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TEST SCHEDULE

Type of Test	Frequency (HZ)	Table Amplitude (inches - peak to peak)
Exploratory	4-33	0.020 ± .004
	34-50	0.006 ± 0.001
Variable Frequency	4-15	0.060 ± 0.012
	16-25	0.040 ± 0.008
	26-33	0.020 ± 0.004
	34-40	0.010 ± 0.002
	41-50	0.006 + 0.000 - 0.001
Endurance	At the resonant frequencies determined from variable frequency test.	At amplitude corresponding to frequency selected for endurance test.

The endurance test of the test Schedule shall be conducted by vibrating the test item for a total period of at least 2 hours at the resonant frequencies chosen by the test engineer based on the Variable Frequency Test results. If no resonant condition results the test shall be performed at 50 HZ. After these vibration tests the test item shall be subjected to the Electrical Tests of 4.5.3.5.

4.6.1.2 Vibration (transportation). The control cable assemblies and connector assemblies, packaged as specified in Section 5.1.1, shall be secured to the vibration table such that the means of securing will not deform the shipping container. The unit in its shipping container shall be vibrated in each of its three mutually perpendicular axes. The vibratory input level shall be maintained at or near the shipping container area of contact with the vibration table. If the input vibration is measured at more than one control point, the control signal shall be the average of all input accelerometers. Sinusoidal cycling shall be conducted at a logarithmic rate for 84 minutes per axis in accordance with the test schedule.

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TEST SCHEDULE

Frequency Range	Amplitude	Sweep Time
5-200HZ Tolerance: ±0.5HZ @<25HZ ±2%@ ≥ 25HZ	1.5gpk±0.15	5-200-5HZ in 12 minutes

After this vibration test, the unit shall be subjected to the tests of 4.5.3.5.

4.6.1.3 Shock (nonoperating). Torpedo control cables and connector plugs and receptacles, mounted in a fixture which simulates the actual mated coupling in service installation (Figure 6 illustrates test setup for Type I Class 1 cable), shall be subjected to shock tests as specified in MIL-S-901 for Grade A, Class I, light weight equipment.

4.6.1.4 Temperature and humidity

4.6.1.4.1 Humidity. The torpedo control cable assemblies and connector plugs and receptacles (all types) shall be placed within a test chamber whose temperature shall be raised to 158 degrees F ± 5 degrees F and maintained for a period of 96 hours at a relative humidity of 95 percent ± 5 percent. At the end of the test period the equipment shall be allowed to return to room temperature after which insulation resistance and continuity tests shall be performed in accordance with 4.5.3.5.

4.6.1.4.2 High temperature test, equipment operating. Cable assemblies and connector plugs and receptacles (all types) shall be placed within a test chamber whose temperature shall be raised to 130 degrees F ± 5 degrees F and sustained for a period of 48 hours. Type I Class 1 cables shall be mounted in a torpedo receptacle as shown in Figure 7 for this test. At the end of this test period, and while still at the test temperature, insulation resistance and continuity tests shall be performed. Type I Class 1 cables shall then be subjected to a hydrostatic pressure test of 700 psi ±20 psi for a time duration of 15 minutes minimum in accordance with 4.5.3.5.

4.6.1.4.3 Low temperature test, equipment operating. Cable assemblies and connector plugs and receptacles (except Class 2) shall be placed within a test chamber whose temperature shall be lowered to -20 degrees F ±5 degrees F and sustained for a period of 48 hours. Type 1 Class 1 cables shall be mounted in a torpedo receptacle as shown in Figure 7 for this test. At the end of this test period, and while still at the test temperature, insulation

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resistance and continuity tests shall be performed in accordance with 4.5.3.5. Type I Class 1 cables shall then be subjected to a hydrostatic pressure test of 700 psi ± 20 psi for a time duration of 15 minutes minimum.

4.6.1.4.4 Low temperature test, equipment operating. Class 2 torpedo control cable assemblies and connector plugs and receptacles shall be placed in a test chamber whose temperature shall be lowered to 28 degrees F +3 degrees F -0 degrees F and maintained for a period of 24 hours. At the end of this test period a continuity test shall be performed in accordance with 4.5.3.5.1.

4.6.1.5 Marine environment tests. Drying, wiping, or baking prior to performance checking shall not be accomplished.

4.6.1.5.1 Simulated sea atmosphere. A laboratory salt spray test shall be performed as specified in MIL-STD-810, Method 509.1.

4.6.2 Pressure hydrostatic

4.6.2.1 Pressure 250 psi (external). Surface vessel torpedo tube door receptacle, Type II Class 3, shall be mounted in a fixture simulating service condition of sealing (see Figure 5) and be subjected to a sustained hydrostatic pressure of 250 psi ±10 psi for a time duration of 15 minutes minimum. Any evidence of water in the cavity behind the insert, excessive water leakage past the door seal, mechanical breakage, or permanent distortion of internal components shall be cause for rejection.

4.6.2.2 Pressure 25 psi (external). Torpedo control cable assemblies Type III Classes 2 and 3, connector assemblies Classes 4 and 5, and stowage rack and test cable assemblies Class 6 shall be installed to sample mating connectors listed in Drawing 783794 and be subjected to a sustained hydrostatic pressure of 25 psi minimum and 30 psi maximum for a period of 10 minutes minimum. Evidence of water leakage past the seals, mechanical breakage, or permanent distortion shall be cause for rejection.

4.6.3 Dielectric strength. The contact-insert material shown on Drawing 783794 shall be subjected to the following dielectric tests:

4.6.3.1 Minimum dielectric strength. The contact-insert material shall be tested in accordance with FED-STD-406 Method 4031 ("Test for specified minimum requirements") and with 3.3.4.3 of this specification. Minimum dielectric strength shall be 325 volts per mil without material breakdown.

4.6.3.2 Dielectric breakdown. The contact-insert material shall be tested in accordance with FED-STD-406 Method 4031 ("Test to breakdown, step-by-step"). Failure to meet the requirements therein shall be cause for rejection.

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4.6.4 Cable separation force, cable cutter type. From each lot of 300 Type I Class 1 cable assemblies being manufactured, and upon completing all other tests, 5 each of the following shall be randomly selected:

(a) Pivot screw

(b) Shear pin

(c) Cutter blade

(d) Lock wire

These parts shall be assembled in accordance with Drawing 1515850 and Figure 7. Loop the cable through cable cutter finger. When a force in tension of not less than 50 pounds nor more than 70 pounds is exerted on the cable in the same plane as the cable cutter and parallel to the centerline of the coverplate, the shear pin shall shear and the cable shall be severed.

4.6.5 Pressure 600 psi (cut cable). The breech door plugs of cables completing the cable separation test of 4.6.4 shall be subjected to a sustained hydrostatic pressure of 600 ±20 psi for a time duration of 15 minutes. Any evidence of water leakage through the breech door plug shall be cause for rejection.

5. PACKAGING

5.1 Preservation and packaging

5.1.1 Level A. In addition to the following, Level A packaging shall be in accordance with MIL-E-17555.

5.1.1.1 Cleaning, drying, and preservation application. Not applicable.

5.1.1.2 Unit packaging. Unit packaging of the control cable assemblies and connector assemblies shall be in accordance with Method 1A-8 of MIL-P-116. The unit package shall contain one each.

5.1.1.3 Intermediate package. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Cleaning, drying, and preservative application. Not applicable.

5.1.3.2 Unit packaging. Unit packaging of the control assemblies and connector assemblies shall be in accordance with Method III of MIL-P-116. The unit package shall contain one each.

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5.1.3.3 Intermediate package. Not applicable.

5.2 Packing

5.2.1 Level A. In addition to the following, Level A packing shall be in accordance with MIL-E-17555.

5.2.1.1 Exterior containers. The control cable assemblies and connector assemblies, packaged as described in 5.1.1.2, shall be packed in a wood box in accordance with PPP-B-621, Class 2 or PPP-B-601, Overseas Type or in a wirebound box in accordance with PPP-B-585, Class 3 (weight limit not to exceed 150 lbs) or in a fiberboard box in accordance with PPP-B-636, Class 2 or PPP-B-591, Overseas Type (weight limit not to exceed 70 lbs).

5.2.2 Level B. In addition to the following, Level B packing shall be in accordance with MIL-E-17555.

5.2.2.1 Exterior containers. The control cable assemblies and connector assemblies, packaged as described in 5.1.1.2, shall be packed in a wood box in accordance with PPP-B-621, Class 1 or PPP-B-601, Domestic Type or in a wirebound box in accordance with PPP-B-585, Class 1 (weight limit not to exceed 150 lbs) or in a fiberboard box in accordance with PPP-B-636, Class 1 or PPP-B-591, Domestic Type (weight limit not to exceed 70 lbs).

5.2.3 Level C. The control cable assemblies and connector assemblies, packaged as described in 5.1.3.2, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Levels A, B, or C cushioning. The unit package or pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to insure the control cable assemblies and connector assemblies will meet all the requirements of Section 3 after being tested in accordance with 4.6.1.2 (Vibration, Transportation). At the time of packaging or packing, the moisture content of the cushioning material shall not exceed 5 percent.

5.4 Marking

5.4.1 Special marking. None.

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5.4.2 Normal markings. In addition to the markings required by contract or order, unit packages, intermediate packages (when used), and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 including serial numbers.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. Torpedo control cable assemblies and connectors covered by this specification are intended for use in coupling (completing) electric circuits of the torpedo fire control system from the ship's junction box in the torpedo room of combatant vessels through a multiconductor circuit breaking switch box, through the torpedo tube, and through the torpedo shell to the various elements of the torpedo control mechanism.

6.1.1 Type I. These torpedo control cable assemblies and connectors are intended for use with torpedoes launched from surface vessels or submarines and serve to couple the fire control circuits from the torpedo afterbody to the sea side (submarine) or the inner side (launchers - surface vessel use) of the torpedo tube.

6.1.2 Type II. These torpedo control cable assemblies and connectors are intended for use with torpedoes launched pneumatically from surface vessels and serve to couple Type I cable assemblies to the fire control circuits from the external or shipboard side of the torpedo launcher to the multiconductor circuit breaking switch box.

6.1.3 Type III. This type covers 4- and 10-contact connector assemblies (plugs and receptacles), 65-contact connector assemblies (plugs and receptacles, switch box and "B" cable), 65-contact conductor cable assemblies ("B" and test), and stowage rack receptacles. These units are intended for use aboard surface vessels or submarines and serve generally to couple either the torpedo tube breech door, the ship's junction box, and associated equipment to the multiconductor circuit breaking switch box or test equipment to torpedoes or Type I cables.

6.1.4 Class 1. This class of equipment, consisting of torpedo control cable assemblies and connectors intended for Type I application as specified in 6.1.1, primarily designed for resistance to underwater service conditions encountered in submarine ejection systems, is also used in surface vessel torpedo ejection systems where, though the environmental conditions are different, the ruggedness and electrical requirements are similar.

6.1.5 Class 2. This class of torpedo control cable assembly and connector, intended for Type III application as specified in 6.1.3, is designed for resistance to salt spray, salt atmosphere, and usage characteristic of submarine service. This application calls for a stronger and more rugged coupling requirement than those found in surface vessel torpedo launching systems.

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6.1.6 Class 3. This class of torpedo control cable assembly and connector, intended for Type II application as specified in 6.1.2, is designed for resistance to marine environment and usages characteristic to surface vessel torpedo launching systems. This application calls for a light durable connector coupling compatible with surface vessel torpedo tube requirements which are less severe than those specified for the submarine torpedo ejection systems.

6.1.7 Class 4. This class of torpedo control cable assembly and connector, intended for Type III application as specified in 6.1.3, is designed for use with the "C" cable in coupling the fire control system from the ship's junction box to the multiconductor circuit breaking switch box. In this class, connector design is governed by the size of the cable and is similar to "B" cable design except for the contact identity. In use, this category is applicable to submarine ejection systems as well as surface vessel torpedo launchers.

6.1.8 Class 5. This class of cable connector assembly serves to couple the ship's power into the torpedo fire control circuits and is designed for Type III application.

6.1.9 Class 6. This class of cable and connector assembly is intended for tests, functional check out, and surveillance checks of the torpedo and torpedo fire control systems in the Type III application.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements

(a) Title, number, and date of this specification, and type and class of cable to be procured.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging.

(d) Required level of packing.

(e) Assigned activity for preproduction and periodic production inspections.

(f) That additional samples required because of failure of any units (s) to pass the prescribed preproduction or periodic production inspections shall be provided and inspected at the expense of the supplier.

(g) That samples required for preproduction and periodic production inspections are not included in the number called for in the contract or order.

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6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID
4.4.1	Failure report	DI-R-4805
4.6	Certified inspection data	DI-T-2072

These data will be delivered as identified on the applicable DID only when specified on DD Form 1423 (Contract Data Requirements List) and incorporated into applicable contracts.

(Copies of data item descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. This level shall be used only when it is definitely known that the item will be held in covered storage in overseas locations for six months or less or in domestic locations for an indefinite period.

For Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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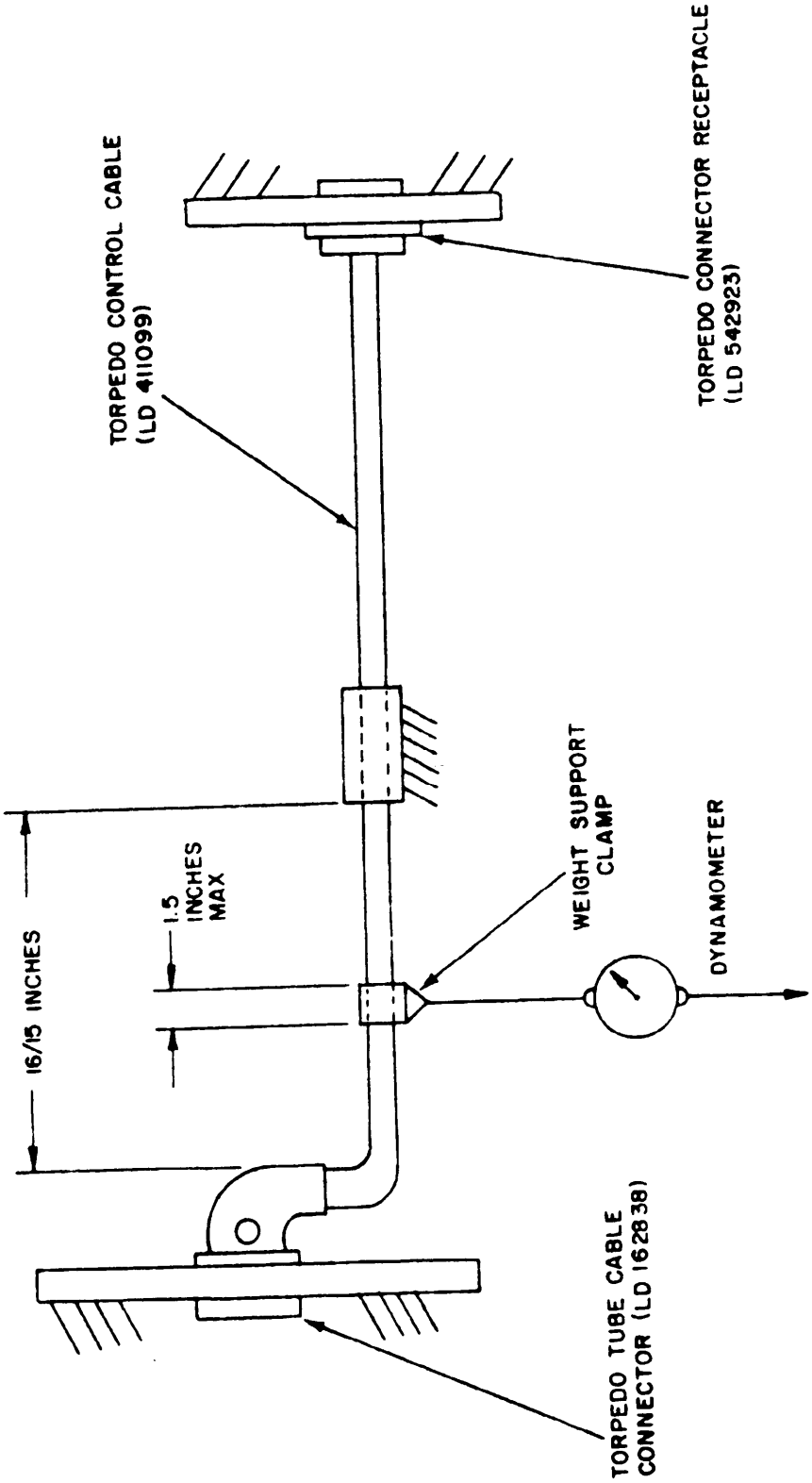


FIGURE 1. TYPICAL TEST SETUP-CABLE ENTRANCE PULL TEST

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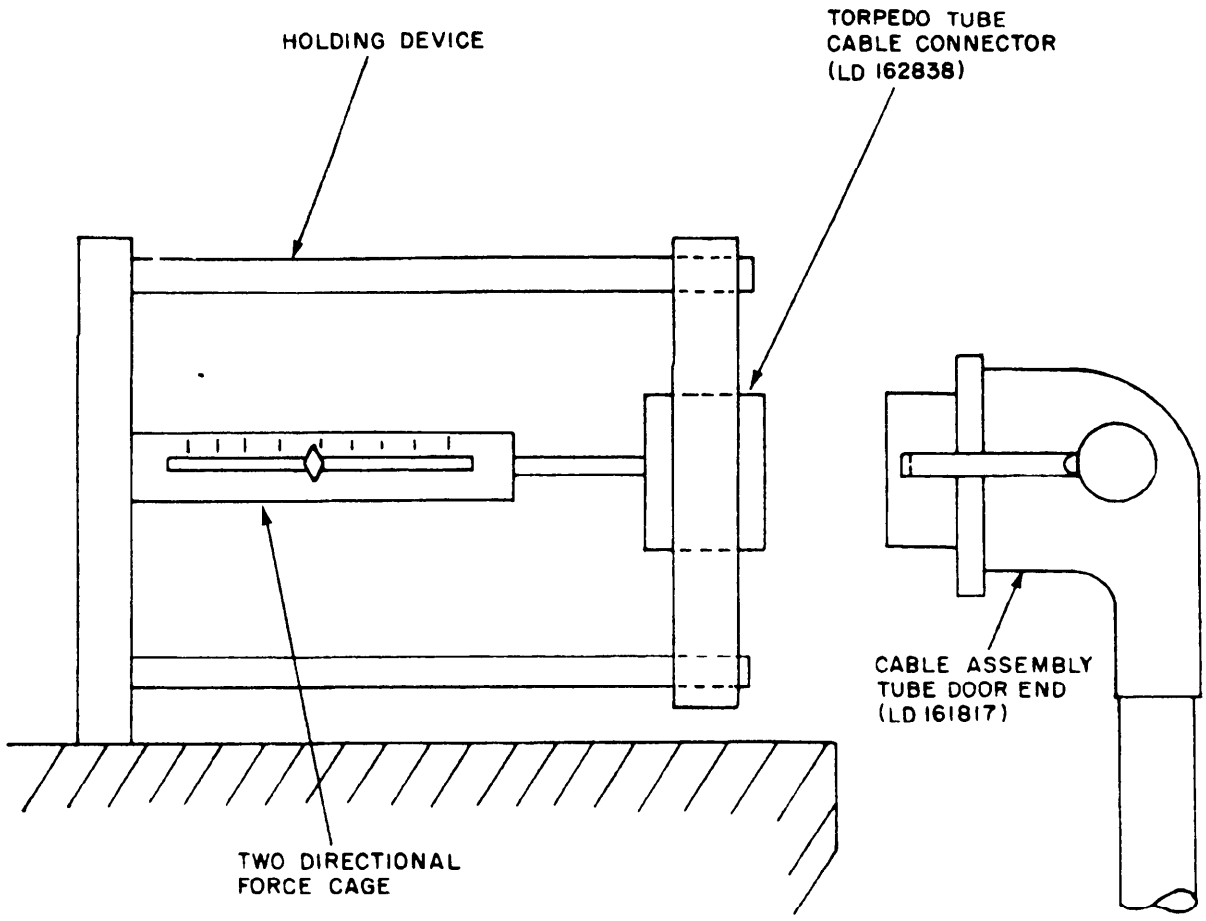


FIGURE 2. TYPICAL TEST SETUP- MATING AND UNMATING FORCE

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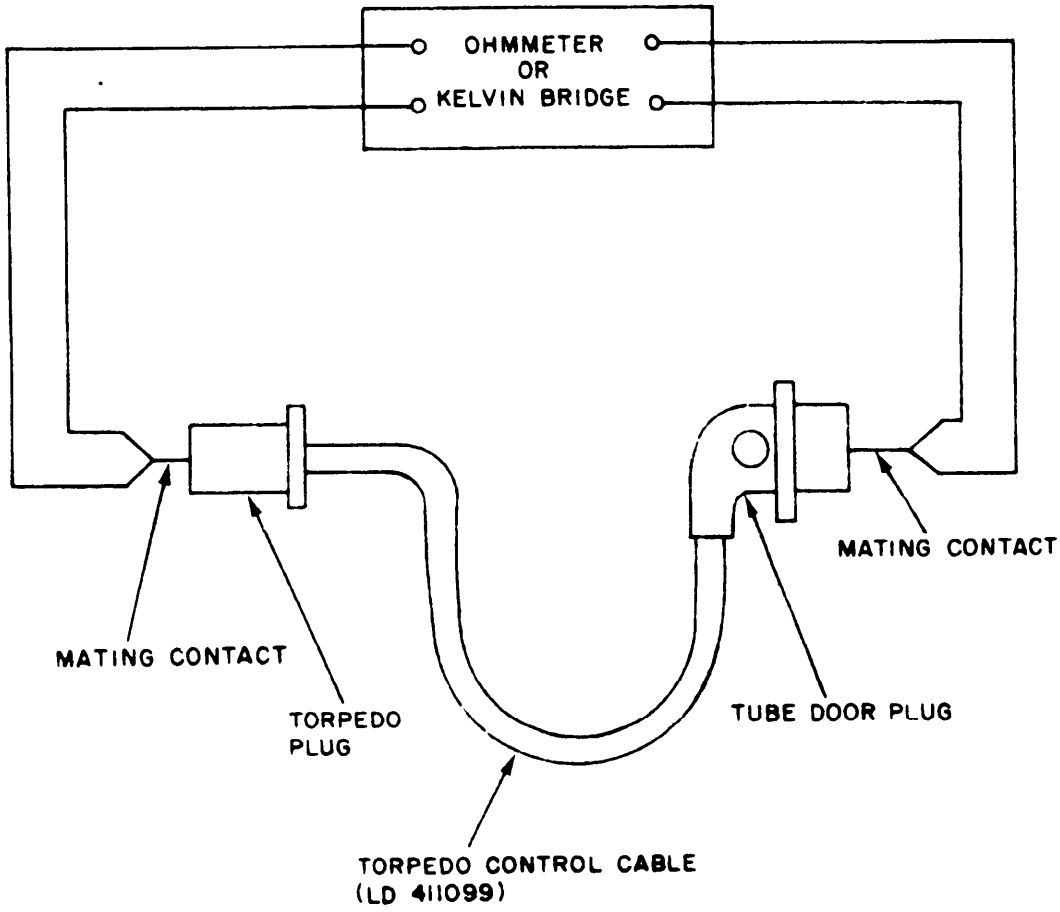


FIGURE 3. TYPICAL TEST SETUP-CONTINUITY TEST

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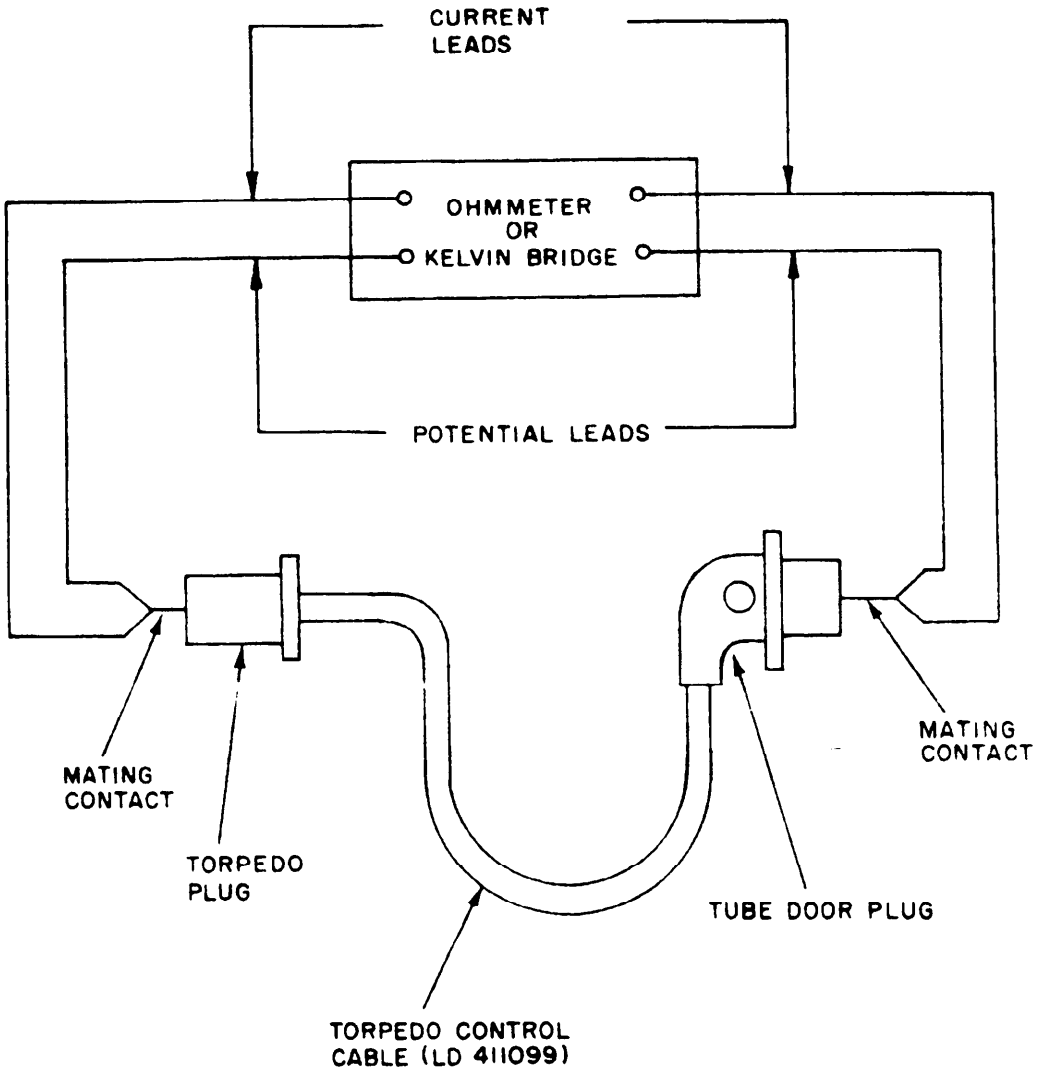


FIGURE 4. TYPICAL TEST SETUP - CONTACT RESISTANCE

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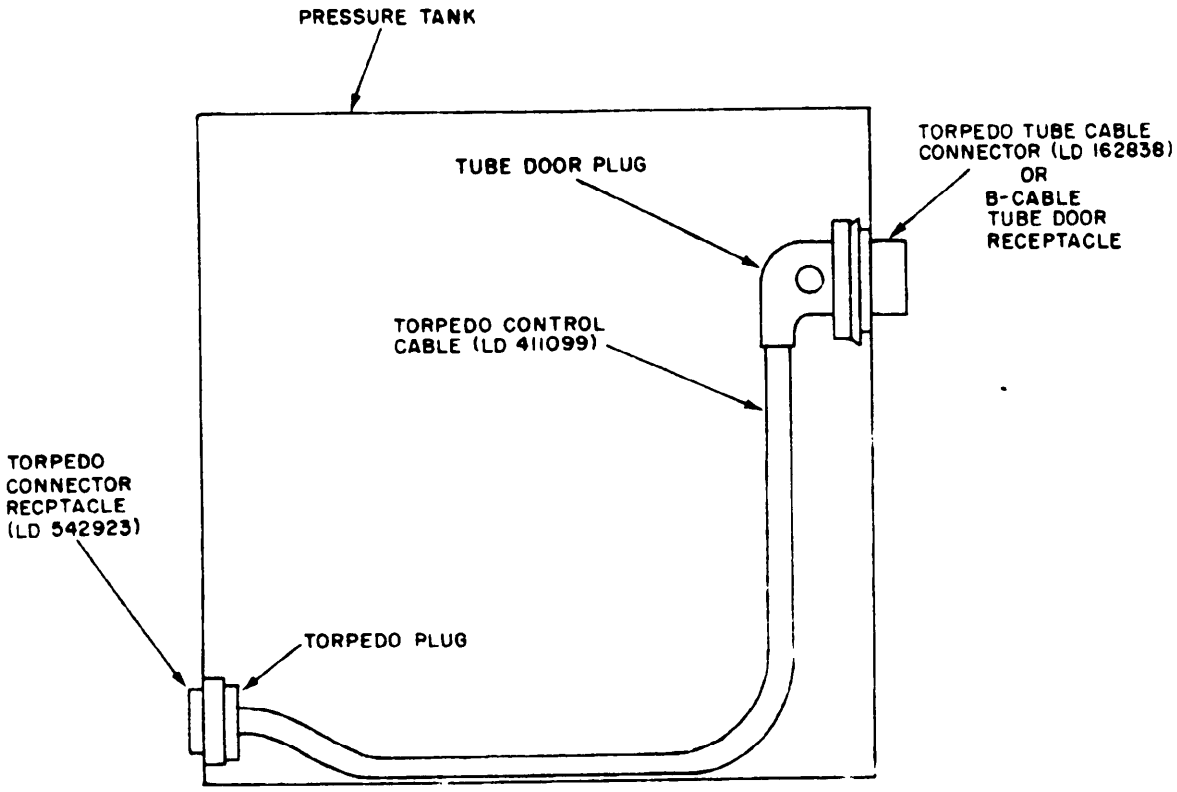


FIGURE 5. TYPICAL TEST SETUP - HYDROSTATIC PRESSURE

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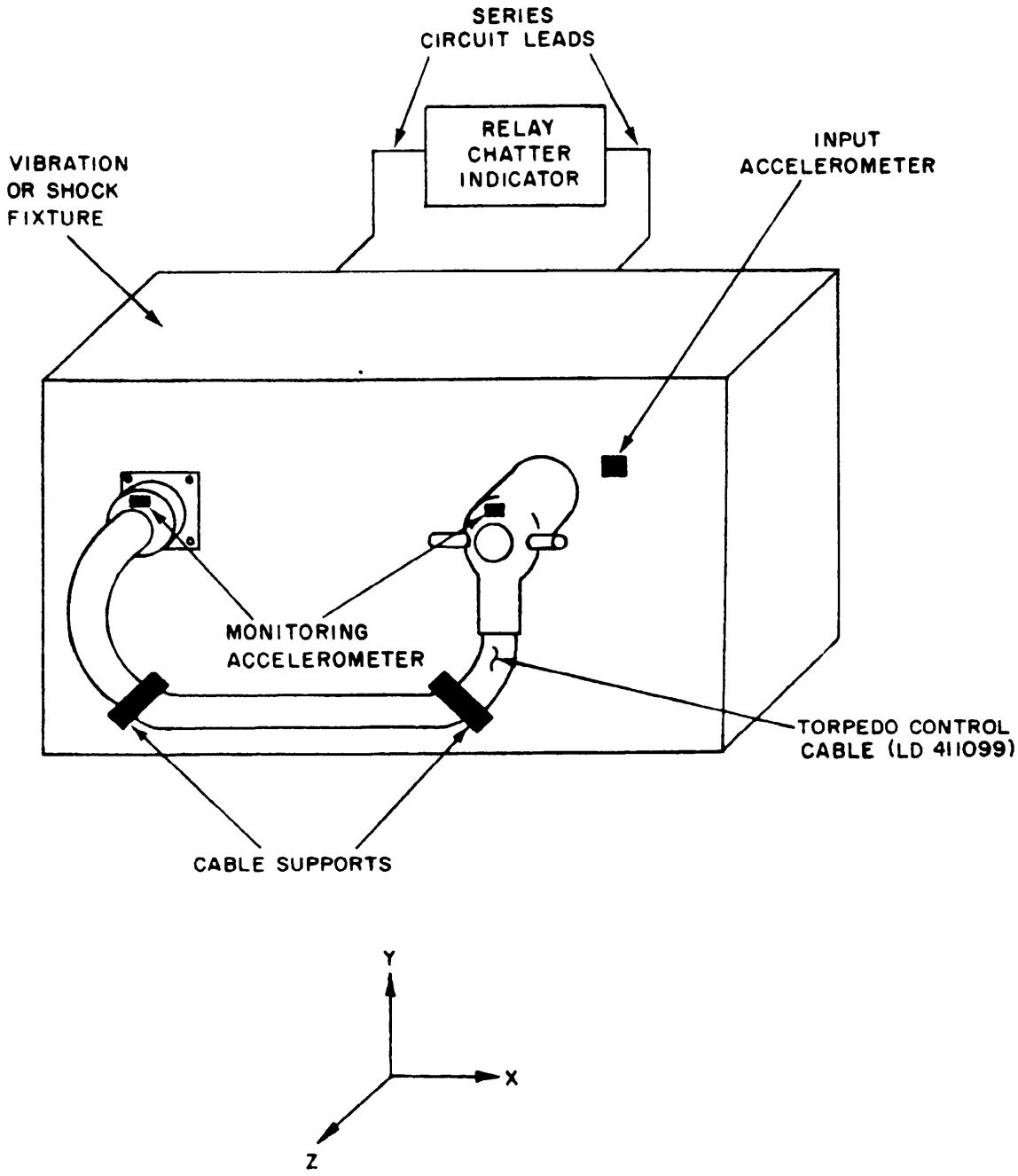


FIGURE 6. TYPICAL TEST SETUP-VIBRATION AND SHOCK

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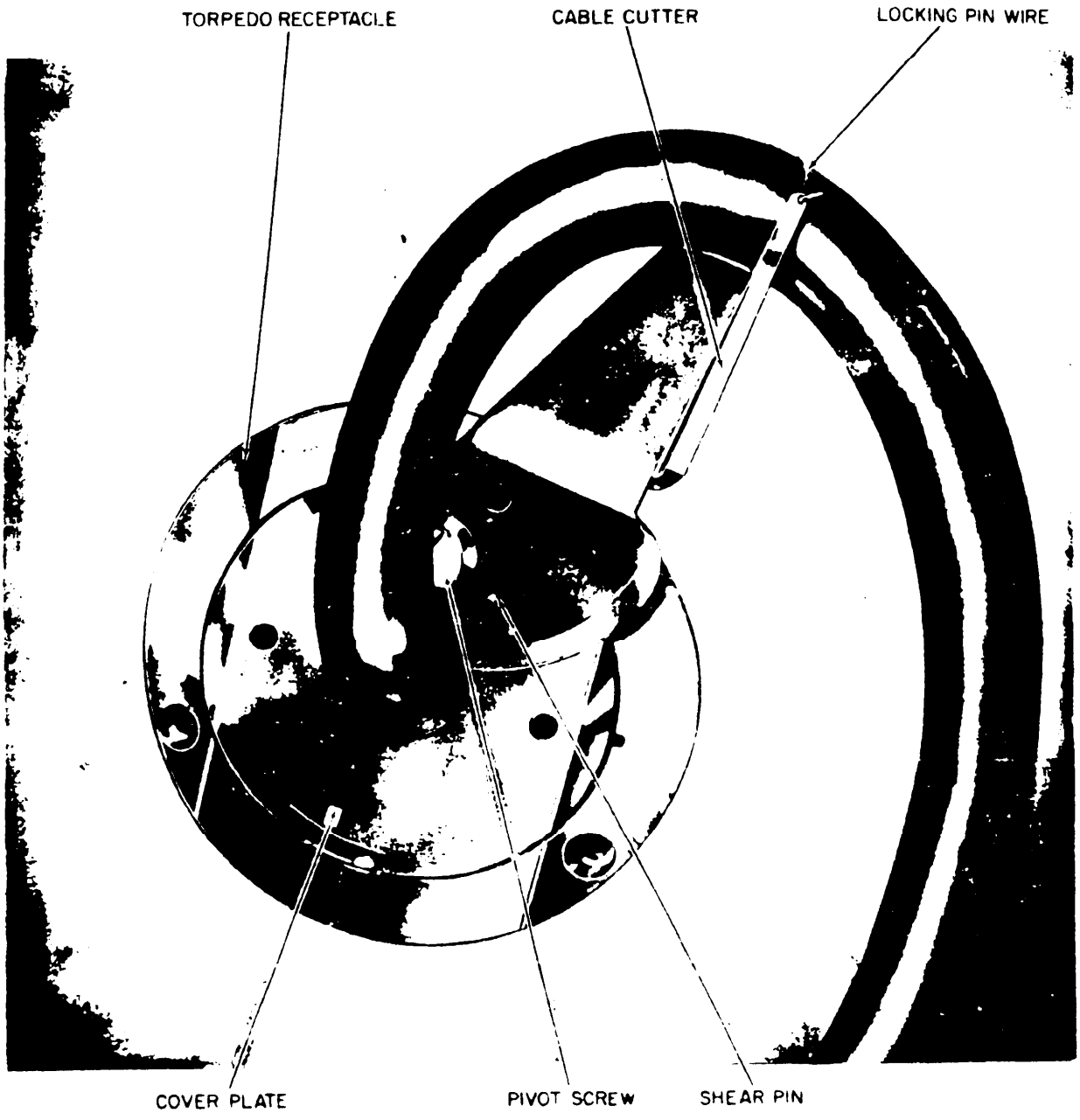


FIGURE 7. MARK 1 A-CABLE WITH CUTTER INSTALLED

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